# IGCSE 0607 Practice Questions — E6.3

## Circles, arcs and sectors

**Total Questions:** 12 — **Total Marks:** 82 Marks

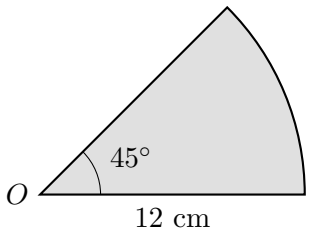
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**1** The diagram shows a sector of a circle, center $O$, with radius 12 cm and sector angle 45°.



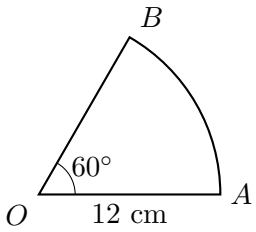

**(a)** Work out the exact length of the arc of the sector.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Work out the exact area of the sector.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**2** The diagram shows a minor sector $AOB$ of a circle with centre $O$ and radius 12 cm.
The sector angle is 60°.



NOT TO SCALE

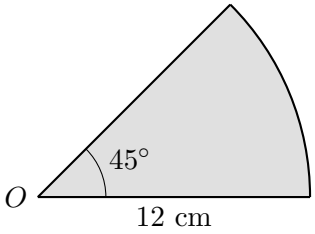
**(a)** Work out the length of the minor arc $AB$.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Work out the area of the minor sector $AOB$.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**3** The diagram shows a sector of a circle with centre $O$ and radius 12 cm.
The sector angle is 45°.



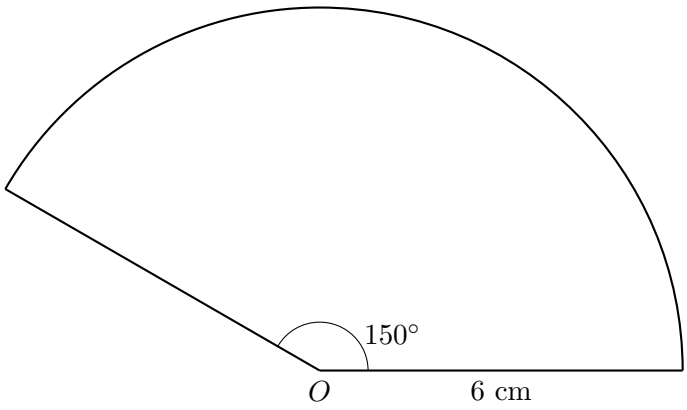

**(a)** Work out the exact area of the sector.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[2]**

**(b)** Work out the exact perimeter of the sector.
Give your answer in terms of $\pi$.

Perimeter = ..... cm **[3]**

**4** The diagram shows a sector of a circle with centre $O$ and radius 6 cm.
The sector angle is 150°.




**(a)** Calculate the area of the sector.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the sector.
Give your answer in terms of $\pi$.

..... cm **[3]**

**5** **(a)** A sector of a circle, center $O$, has radius $r$ cm and sector angle $\theta°$.
The area of the sector is $10\pi$ cm$^2$ and the length of its arc is $2\pi$ cm.

Find the value of $r$ and the value of $\theta$.

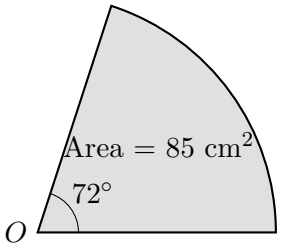
$r =$ ..... , $\theta =$ ..... **[5]**

**(b)** Another sector of a circle has radius $x$ cm and sector angle $(x + 20)°$.
The length of its arc is $\frac{16\pi}{5}$ cm.

Find the value of $x$.
You must show all your working.

$x =$ ..... **[4]**

**6** The diagram shows a sector of a circle with centre $O$ and angle 72°.
The area of the sector is 85 $cm^2$.




NOT TO SCALE

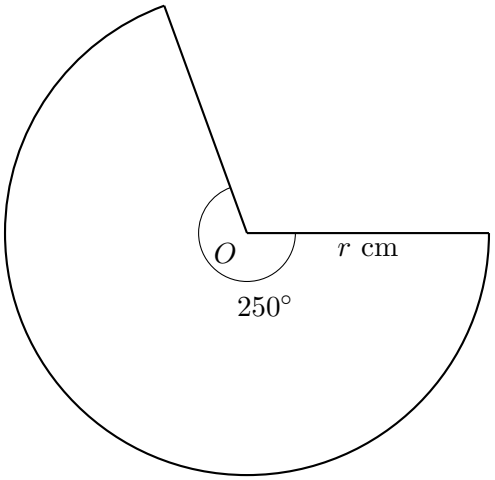
**(a)** Calculate the radius of the sector.

..... cm **[3]**

**(b)** Calculate the perimeter of the sector.

..... cm **[3]**

**7** The diagram shows a major sector of a circle with centre $O$ and radius $r$ cm.
The reflex angle is 250°.
The length of the major arc is 35 cm.



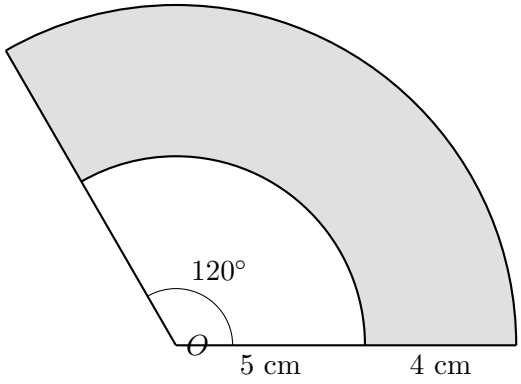

**(a)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(b)** Calculate the area of the major sector.

..... $cm^2$ **[3]**

**8** The diagram shows a shaded shape made from two concentric sectors of circles with centre $O$.
The inner radius is 5 cm and the outer radius is 9 cm.
The angle of both sectors is 120°.




**(a)** Calculate the area of the shaded shape.

Area = ..... $cm^2$ **[3]**

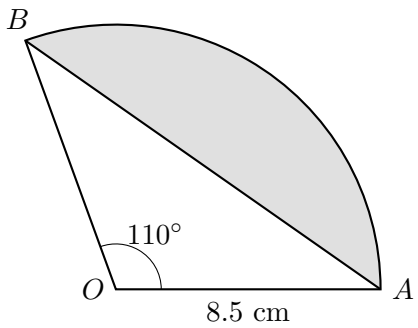
**(b)** Calculate the perimeter of the shaded shape.

Perimeter = ..... cm **[3]**

**(c)** A complete circle has the same area as the shaded shape.
Calculate the radius of this circle.

Radius = ..... cm **[2]**

**9** The diagram shows a sector $OAB$ of a circle with center $O$ and radius 8.5 cm.
The sector angle is 110°.
A straight line connects $A$ and $B$, forming a triangle $OAB$ and a shaded segment.




**(a)** Calculate the area of the sector $OAB$.

..... $cm^2$ **[2]**

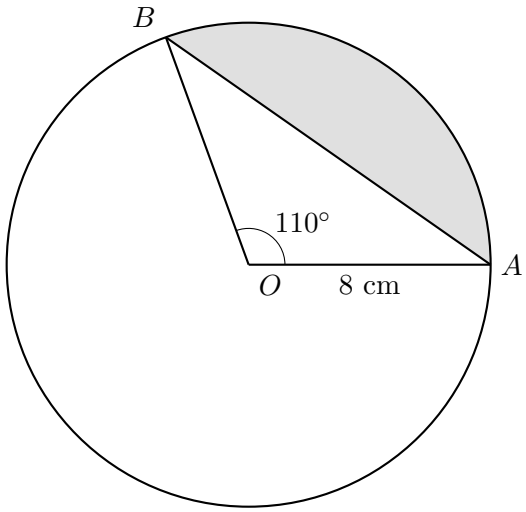
**(b)** Calculate the area of the shaded segment.

..... $cm^2$ **[3]**

**(c)** Calculate the perimeter of the shaded segment.

..... cm **[4]**

**10** The diagram shows a circle with centre $O$ and radius 8 cm.
$A$ and $B$ are points on the circumference such that angle $AOB = 110°$.




**(a)** Calculate the area of the shaded segment.

..... $cm^2$ **[4]**

**(b)** Calculate the perimeter of the shaded segment.

..... cm **[3]**

**11** A sector of a circle has radius $r$ cm and sector angle $x°$.

**(a)** The arc length of the sector is $6\pi$ cm and the area of the sector is $27\pi$ cm$^2$.

**(i)** Work out the value of $r$.

$r =$ ..... **[3]**

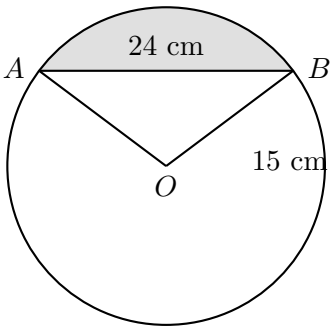
**(ii)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** Another sector has radius $R$ cm and sector angle $y°$.
The perimeter of this sector is equal to the circumference of a circle with radius $R$ cm.
Find an expression for $y$ in terms of $\pi$.

$y =$ ..... **[4]**

**12** The diagram shows a circle, centre $O$, with radius 15 cm.
$AB$ is a chord of the circle with length 24 cm.
The region bounded by the chord $AB$ and the minor arc $AB$ is shaded.




NOT TO SCALE

**(a)** Show that angle $AOB = 106.3°$, correct to 1 decimal place.

**[3]**

**(b)** For the following calculations, use angle $AOB = 106.3°$.

**(i)** Calculate the perimeter of the shaded minor segment.

..... cm **[3]**

**(ii)** Calculate the area of the shaded minor segment.

..... $cm^2$ **[4]**